Dr. Vivek Mahato, PhD

Postgraduate Researcher

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Work Experience

2021/04 – present
Dublin, Ireland

Postgraduate Researcher
I-Form Advanced Manufacturing Research Centre, Dublin City University

- Working on real-time process monitoring and control, and parameter optimisation and recommendation for Additive Manufacturing (AM) process.
- Analysing sensor data in real-time to anticipate defects and alert the operator.
- Employing Machine Learning in AM to maintain higher quality products while reducing energy cost and waste significantly.
- Analysing the efficiency and robustness of the employed algorithms when adapted in the setting of different parameters and materials.

2021/04 – present

Researcher
Whyze Health

- Employing statistics and Machine Learning to guide health patients to the right care.
- Provide clinic officials key insights and post-treatment quality-of-life metrics of the patients.

2017/07 – 2017/08
Dublin, Ireland

Business Analyst Intern
Investment Technology Group

- Developed application to automatically generate report of daily transactions of the company, which then is parsed to MHTML format for distribution through emails.
- Assisted the BA team by monitoring live SQL databases and dashboard creation with Power BI.

2014/09 – 2016/07
Kolkata, India

Web Developer
Enigma Infosolutions Pvt. Ltd

- Developed Web Applications using HTML, JavaScript, and PHP.
- Assisted with Network Administration and Software Development.
- Built and maintained databases using MySQL, and Oracle.

Education

2017/09 – 2021/05
Dublin, Ireland

PhD in Machine Learning
University College Dublin
- Dissertation: Real-Time Machine Learning in Manufacturing
- Supervisor: Prof. Pádraig Cunningham
  - Research and Development of Machine Learning techniques for predictive control over Additive Manufacturing (using PBF-LB) process.
  - Designed a two-stage classification strategy to speed up time series classification with the least compromise on accuracy (than the state-of-the-art models).
  - Achieved Machine Learning assessment faster than the manufacturing process.

2016/09 – 2017/08
Dublin, Ireland

Master's in Computer Science
University College Dublin
- Major: Data Science
- Dissertation: Neural Embeddings for Automatic Sentiment Analysis (NLP)
- Core Modules:
  - Advanced Machine Learning
  - Text Analytics
  - Big Data Programming
  - Information Visualization

2011/08 – 2014/09
Kolkata, India

Bachelor's in Computer Applications
Techno India Salt Lake, WBUT
- Major: Software Development
- Core Modules:
  - Advanced Data-Structures in Java
  - Computer Programming (Java, C/C++, ASP.NET)
  - Database Management System (DBMS)

Publications

2021
Speedup Methods for k-Nearest Neighbour Dynamic Time Warping
Expert Systems with Applications
(Under Review)

2020
Detecting Pores in 3D Printing Using Melt Pool Time Series Data
Journal of Intelligent Manufacturing

2020
An Evaluation of Classification Methods for 3D Printing Time-Series Data
21st International Federation of Automatic Control : World Congress publication description

2020
Using Navigable Small Worlds to Speed Up Time-Series Classification
28th Irish Conference on Artificial Intelligence and Cognitive Science

2019
Scoring Performance on the Y-Balance Test
The International Conference on Case-Based Reasoning (ICCBR)

2018
A Case-Study on the Impact of Dynamic Time Warping in Time Series Regression
Advanced Analytics and Learning over Temporal Data : ECMLPKDD

2018
A Comparison of k-NN Methods for Time Series Classification and Regression
26th AIAI Irish Conference on Artificial Intelligence and Cognitive Science

Skills

Programming
Python, R, Java

Toolkits
Pandas, Numpy, Scikit-learn, Tensorflow, Pytorch, Flask

Database Technologies
SQL, Hadoop, Spark

Soft Skills
Communication, Team Player, Leader, Problem Solving, Analytical

Machiner Learning
Time-Series, Classification, Regression, NLP, Supervised and Unsupervised Learning

Visualisation
Matplotlib, Seaborn, Tableau, Plotly, D3.js, PowerBI, Qlik

Cloud Systems
AWS, Azure